TECHNICAL SALES ANALYSIS

Customer Sales Performance Analysis

An exploratory assessment of product, temporal, geographic, and order trends

PRODUCT	TIME	GEOGRAPHY	ORDERS
Category & sub-category	Annual & monthly trends	State & city performance	Demand patterns

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DOI: 10.5281/zenodo.24132634

Technical white paper • 2026

Executive Summary

This white paper presents an exploratory analysis of customer sales data found in sales.csv, a secondary dataset obtained from Kaggle. The analysis is done with respect to aspects of sales performance: - Performance of products and categories - Performance for the year - Geographic distribution - Monthly order activity. From the source analysis, Furniture is the leading category, with tables as a leading product. The annual trend visual depicts the maximum number of profits ever recorded in 2022 and a steep decline by 2025. Geographically, Florida and New York are the states with the most sales, with Ohio reporting the least. The monthly order trend shows considerable variation in order quantity throughout the study period. However, the source narrative does not specify the exact calendar month of the maximum.

ANALYTICAL CONTRIBUTION

The value of this study is its organization of sales information into a reusable decision-support structure: product performance, time-based performance, geographic concentration, and order-demand patterns. The framework can help users translate descriptive sales data into practical decisions on inventory, marketing, customer engagement, and market coverage.

Study at a Glance

Data source	sales.csv; secondary data obtained from Kaggle
Analysis scope	Product, annual, geographic, and monthly order performance
Study period shown	2020–2025 in the supplied visualizations; selected narrative sections emphasize 2022–2025
Primary analytical mode	Descriptive and comparative analysis
Intended use	Sales planning, marketing, inventory planning, customer engagement, and market expansion

1. Introduction

Customer sales analysis provides a structured way to examine what customers purchase, when demand changes, where sales are concentrated, and which products contribute most strongly to business performance. This study applies that perspective to a secondary sales dataset and organizes the findings around four practical research questions. The purpose of the analysis is to identify observable sales patterns that can inform business decisions. Rather than treating individual transactions in isolation, the analysis groups the available information into product categories, time periods, geographic locations, and order activity. This makes the findings easier to interpret and apply to inventory planning, marketing, customer relationship management, and geographic expansion.

Research Questions

01	Which product generates the highest sales revenue?
02	In which year did the business record the highest profit?
03	Which city and state have the highest number of customers and generate the highest sales?
04	In which month was the highest number of orders placed?

Study Boundaries

The white paper is intentionally grounded in the supplied source document. Where the source provides a finding, that finding is reported. Where the source narrative does not provide a specific value or month, this paper does not invent one.

2. Data and Analytical Approach

The analysis uses sales.csv, described in the source document as secondary data obtained from Kaggle. The supplied analysis organizes the dataset into four analytical views: product/category performance, annual performance, state and city performance, and monthly order trends.

Analytical Workflow

01	02	03	04
SOURCE	**GROUP**	**COMPARE**	**APPLY**
sales.csv	Product/time/place	Performance patterns	Business decisions

The supplied visualizations provide the principal evidence for the reported comparisons. Because the underlying CSV was not supplied separately, this white paper does not claim additional statistical tests, causal relationships, forecasting accuracy, or data-quality procedures beyond what is visible in the source material.

Interpretive Principle

DESCRIPTIVE, NOT CAUSAL

Observed peaks and declines are treated as patterns in the dataset. The recommendations identify practical areas for management attention, but the analysis does not establish that marketing, inventory, customer behavior, or other factors caused the observed changes.

3. Finding I: Product and Category Performance

The products are classified into three major categories: Electronics, Furniture, and Office Supplies. The supplied category/sub-category visualization shows Tables as the strongest individual sub-category within the Furniture group, while the category-level visualization shows Furniture as the leading category by quantity.



Figure 1. Product-category and sub-category quantity profile reproduced from the supplied analysis.

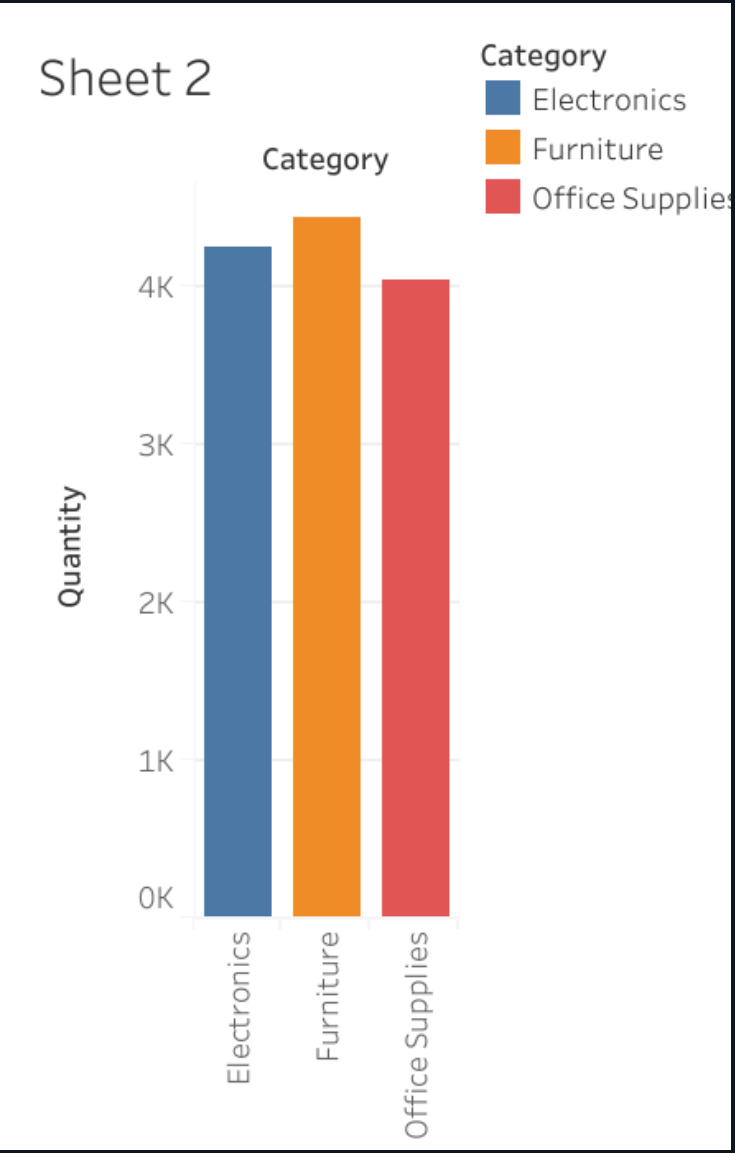


Figure 2. Category-level quantity comparison reproduced from the supplied analysis.

KEY FINDING

Furniture, particularly Tables, is identified in the source analysis as a top-performing segment and a major contributor to overall sales performance.

Business Implication

The source analysis recommends maintaining adequate stock levels, improving product availability, and developing targeted marketing strategies around strong-performing products. At the same time, the performance of other products can be reviewed to identify opportunities for improvement and growth.

4. Finding II: Annual Performance Trend

The annual performance visualization covers 2020–2025 and reports a measure labelled "Count of Profit." The highest point occurs in 2022, after which the measure declines, with a particularly pronounced decrease in 2025.

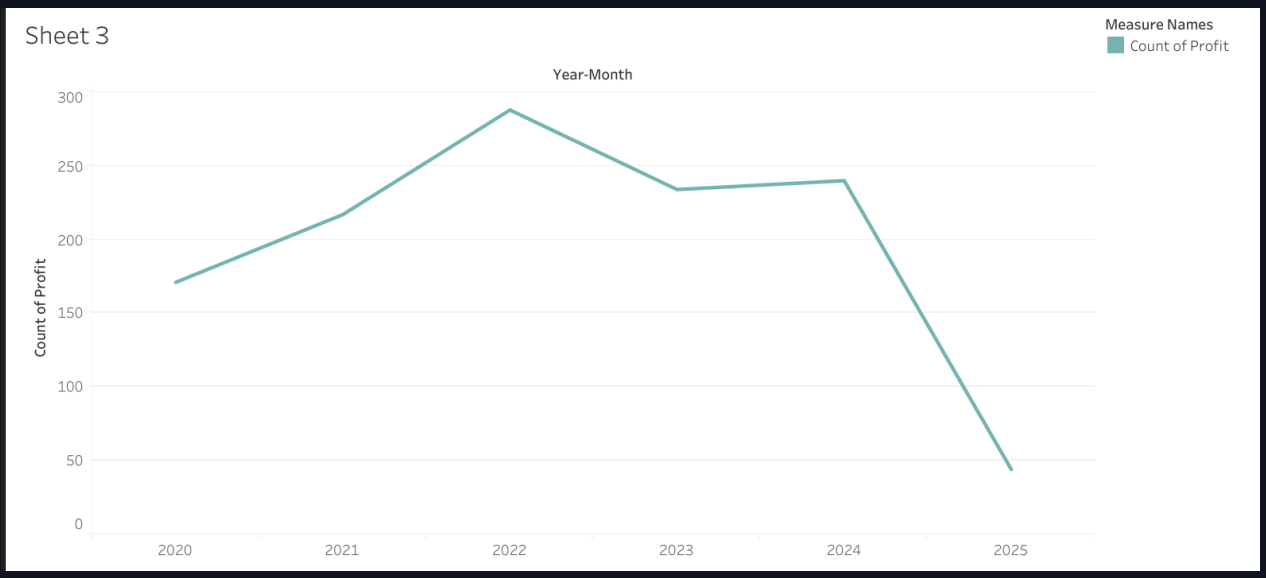


Figure 3. Annual profit-count trend, 2020–2025, as presented in the supplied analysis.

HIGHEST ANNUAL POINT

2022 records the highest point in the supplied profit-count visualization. The source narrative describes this more generally as the peak in overall sales performance.

Interpretive Note

The research question refers to the year with the highest profit, while the supplied figure is labelled "Count of Profit" and the accompanying narrative refers to sales performance. Accordingly, this paper reports 2022 as the highest annual point shown in the source evidence, without converting the count into a monetary profit value.

Management Implication

The source analysis recommends stronger investment in marketing and sales activities, improved customer engagement and brand awareness, more effective sales strategies, stronger customer relationship management, and investigation of the factors associated with the decline.

5. Finding III: Geographic Sales Performance

The geographic analysis compares sales performance across states and cities and identifies locations that contribute materially to overall business revenue. The supplied visualization covers Florida, New York, California, Texas, Illinois, and Ohio, with city-level segments shown within each state.

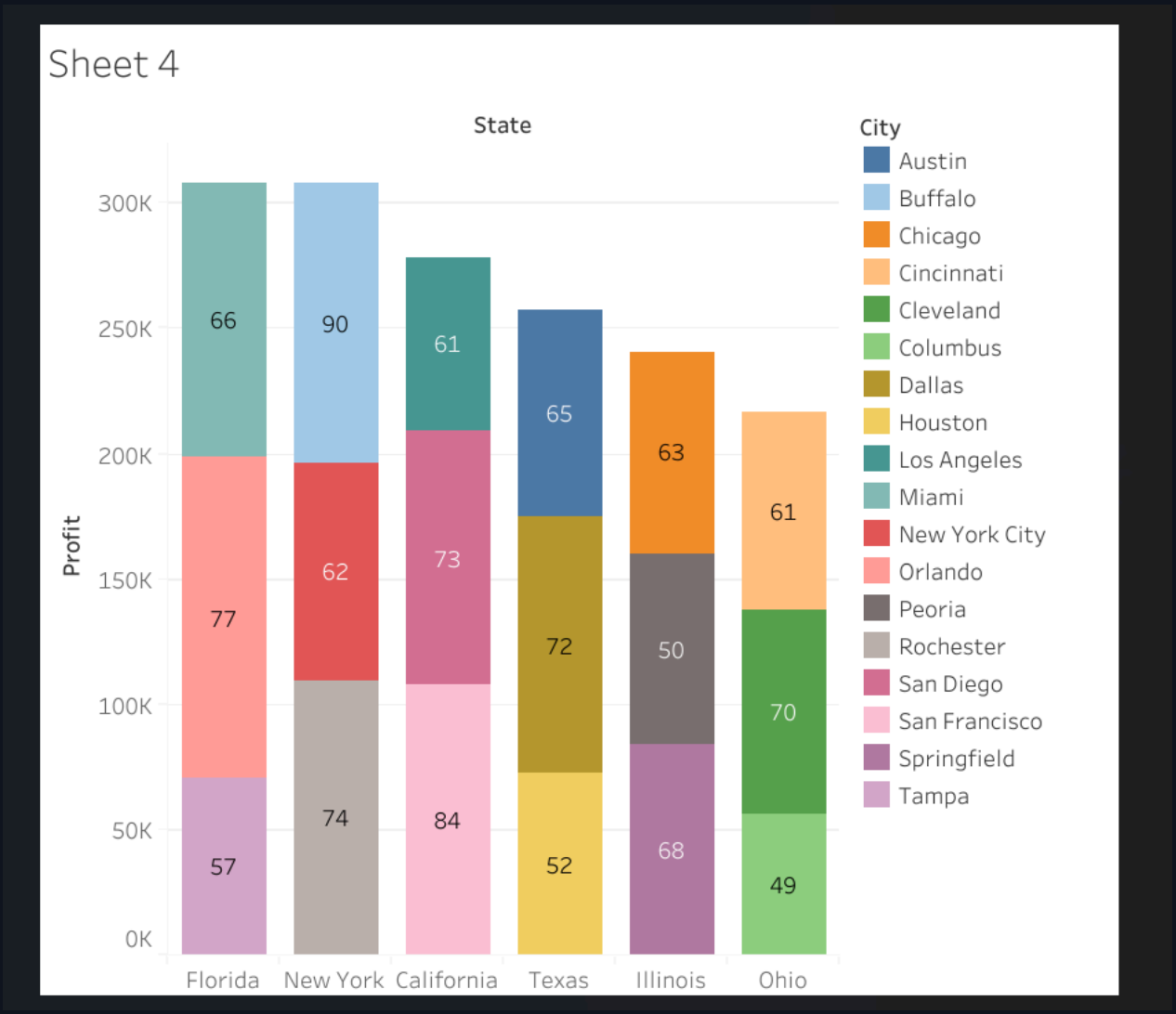


Figure 4. State and city performance visualization reproduced from the supplied analysis.

GEOGRAPHIC FINDING

The source narrative reports that Florida and New York have the most sales, while Ohio has the least.

Business Implication

The source analysis recommends strengthening the business presence in high-performing locations while directing additional marketing, promotional, and brand-awareness activity toward lower-performing areas. This approach is intended to improve geographic coverage, customer awareness, order volume, and growth opportunities.

Decision Lens

Performance signal	Management focus	Expected business use
High-sales locations	Maintain and strengthen presence	Protect established demand
Lower-sales locations	Increase targeted promotion	Expand market coverage
City-level variation	Review local performance	Prioritize customer and marketing effort

6. Finding IV: Monthly Order Trends

The monthly order visualization shows substantial variation in order quantity across the period under review. It contains repeated peaks and troughs, indicating that customer ordering activity is not uniform across months. The source narrative states that the visualization identifies the months with the highest number of orders, but it does not name the specific calendar month in the accompanying text.

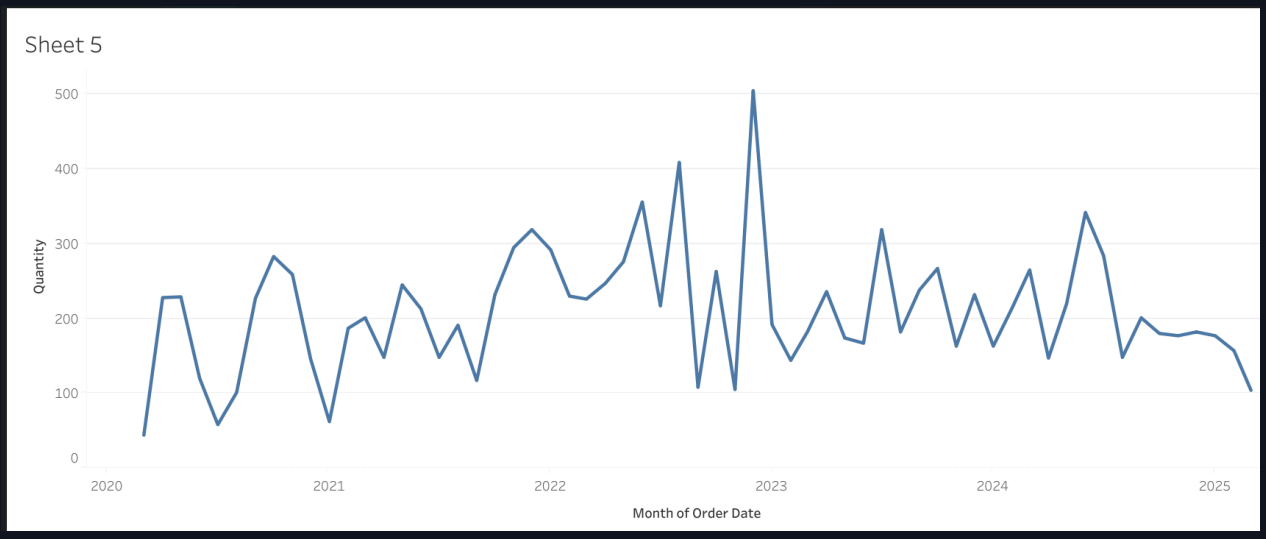


Figure 5. Monthly order-quantity trend reproduced from the supplied analysis.

EVIDENCE BOUNDARY

The source material supports a finding of substantial monthly variation and identifies the existence of peak periods, but it does not provide the exact peak month in text. The paper therefore does not assign a calendar month without the underlying date-level data.

Business Implication

The source analysis recommends using high-order periods as strategic windows for targeted marketing campaigns, promotional offers, and improved inventory planning. Management should also examine the causes of the broader decline observed in 2025.

7. Proposed Analytical Contribution and Transferability

The study contributes a compact sales-analysis structure that connects four commonly separated business questions: what sells, when performance changes, where sales are concentrated, and when customer orders intensify. Organizing these dimensions together makes the analysis more useful for decision-making than reviewing individual charts without a common framework.

Reusable Analytical Structure

Product	Time	Geography	Demand
Top categories	Peak year	Top states/cities	Peak periods
Stock priorities	Trend changes	Market coverage	Campaign timing
Product strategy	Performance monitoring	Expansion focus	Inventory planning

Dissemination and Use Beyond the Original Analysis

For Zenodo publication, the white paper can function as a reusable technical record of the analytical workflow and its findings. The structure is suitable for other analysts, students, business intelligence practitioners, and organizations working with comparable sales datasets. The figures and narrative provide a compact reference for reproducing the analytical questions and translating descriptive results into practical business actions. Because the source material does not document a formal external deployment, partnership, or institutional adoption, this paper does not claim that the analysis has already been adopted outside its originating context. Its broader value is presented as transferability of the analytical structure rather than demonstrated external impact.

8. Recommendations

01	**Prioritize strong-performing products** Maintain adequate stock, improve product availability, and support leading products with targeted marketing.
02	**Respond to the annual decline** Strengthen sales and marketing activities, improve customer engagement, and investigate factors associated with the decline after 2022.
03	**Protect high-performing markets** Maintain strong presence in Florida and New York while using geographic analysis to identify expansion opportunities.
04	**Target lower-performing markets** Use promotional campaigns and brand-awareness initiatives to improve customer acquisition and order volume in weaker locations.
05	**Plan around demand peaks** Use high-order periods for campaign timing, promotional offers, and inventory planning.
06	**Strengthen monitoring** Repeat the four-part analysis periodically so management can track product, time, location, and demand changes.

9. Limitations

- The analysis is based on secondary data described as having been obtained from Kaggle.
- The underlying CSV was not separately available for this white-paper production; therefore, the paper relies on the findings and visualizations contained in the supplied document.
- The source material does not provide monetary profit values, even though one research question refers to profit.
- The source narrative does not name the exact calendar month with the highest number of orders.
- The analysis is descriptive and does not establish causal explanations for observed increases or declines.

10. Conclusion

This study provides a structured view of customer sales performance across product, time, geography, and order activity. The supplied evidence identifies Furniture and particularly Tables as strong-performing segments; 2022 as the highest annual point in the supplied profit-count visualization; Florida and New York as leading sales states; and substantial variation in monthly order activity.

The findings support practical actions in inventory planning, marketing, customer engagement, geographic expansion, and performance monitoring. Most importantly, the analysis demonstrates how a focused descriptive framework can turn sales records into a coherent set of management questions and decision points while maintaining a clear boundary between observed patterns and unsupported causal claims.

ZENODO-READY POSITIONING

This white paper is presented as a technical and exploratory data-analysis record. Its contribution is the documented analytical structure and interpretation of the supplied sales evidence, not a claim of causal inference, validated forecasting, or external deployment.

Source Note

Primary analytical source: Adomi Ogaga Benjamin, “sales.csv, secondary data gotten from Kaggle,” and the accompanying customer sales analysis supplied for this white paper. Figures in this document are reproduced from the supplied analysis.

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